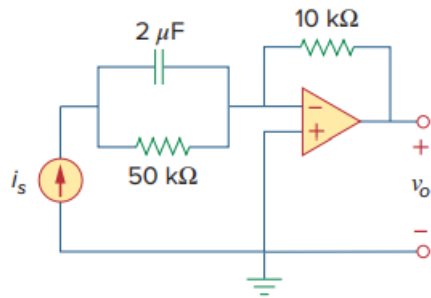


Problem

Determine $v_o(t)$ for $t > 0$ in the circuit of Fig. 7.139. Let $i_s = 10u(t) \mu\text{A}$ and assume that the capacitor is initially uncharged.

Fig. 7.139:



Step-by-step solution

Step 1 of 5

Refer to Figure 7.139 in the text book.

For $t < 0$, the current source value is, $i_s = 0 \text{ A}$

Draw the circuit diagram for $t < 0$.

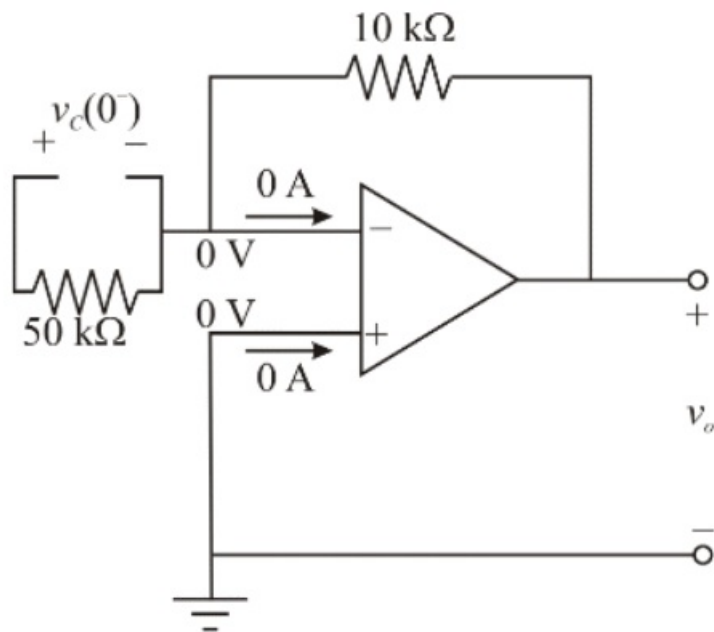


Figure 1

Step 2 of 5

The initial voltage across the capacitor is,

$$v_c(0^-) = 0 \text{ V}$$

Draw the circuit diagram for $t > 0$.

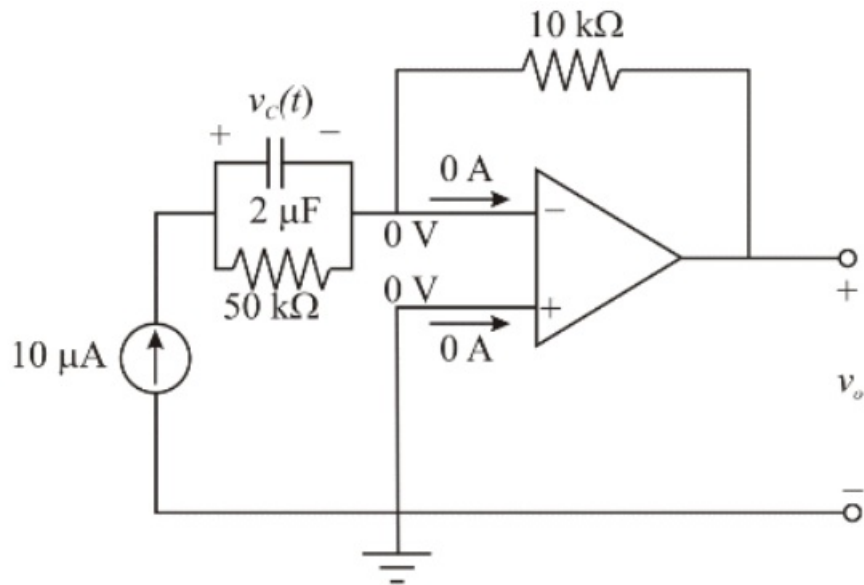


Figure 2

Determine the value of time constant.

$$\begin{aligned} \tau &= RC \\ &= (50 \times 10^3)(2 \times 10^{-6}) \\ &= 0.1 \text{ s} \end{aligned}$$

Step 3 of 5

Draw the circuit diagram for $t = \infty$.

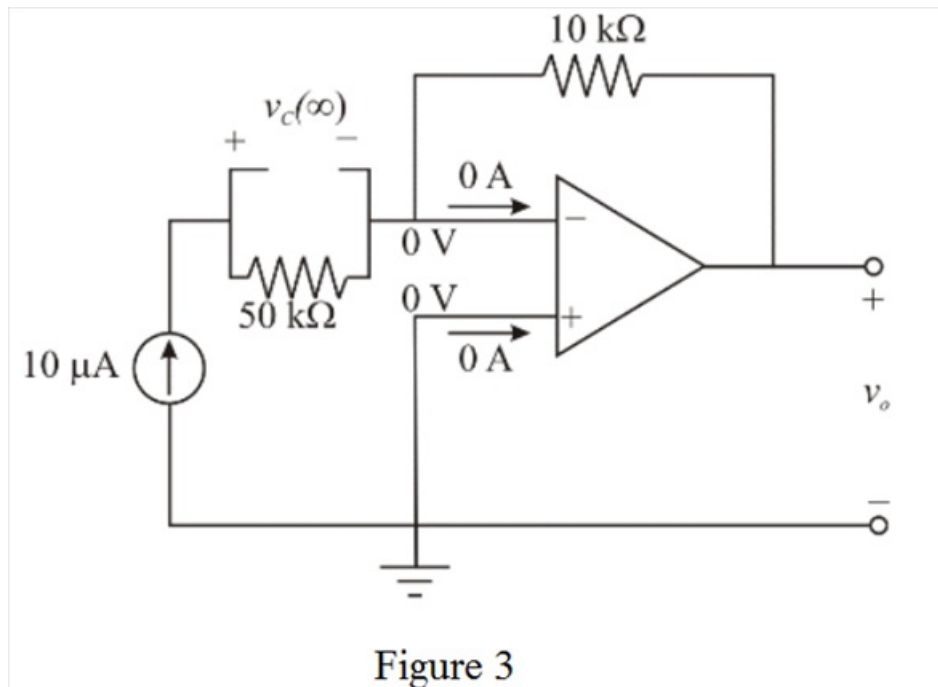


Figure 3

Step 4 of 5

Determine the voltage across the capacitor at $t = \infty$.

$$\begin{aligned}v_c(\infty) &= (50 \text{ k}\Omega)(10 \mu\text{A}) \\&= (50 \times 10^3)(10 \times 10^{-6}) \\&= 0.5 \text{ V}\end{aligned}$$

Determine the voltage across the capacitor.

$$\begin{aligned}v_c(t) &= v_c(\infty) + [v_c(0) - v_c(\infty)]e^{\frac{-t}{0.1}} \\&= 0.5 + [0 - 0.5]e^{\frac{-10t}{1}} \\&= 0.5(1 - e^{-10t})\end{aligned}$$

Step 5 of 5

Write the differential equation in v_c .

$$i_s = C \frac{dv_c}{dt} + \frac{v_c}{50 \text{ k}\Omega}$$

Write the expression for the output voltage.

$$v_o = -i_s(10 \text{ k}\Omega)$$

Substitute $C \frac{dv_c}{dt} + \frac{v_c}{50 \text{ k}\Omega}$ for i_s .

$$v_o = -\left(C \frac{dv_c}{dt} + \frac{v_c}{50 \text{ k}\Omega}\right)(10 \text{ k}\Omega)$$

Substitute $0.5(1 - e^{-10t})$ for v_c and $2 \mu\text{F}$ for C .

$$\begin{aligned}v_o(t) &= -\left\{(2 \times 10^{-6}) \frac{d}{dt}[0.5(1 - e^{-10t})] + \frac{0.5(1 - e^{-10t})}{50 \text{ k}\Omega}\right\}(10 \text{ k}) \\&= -\left[(2 \times 10^{-6})(0.5e^{-10t}) + 10^{-5}(1 - e^{-10t})\right](10 \text{ k}) \\&= -(10^{-5}e^{-10t} + 10^{-5} - 10^{-5}e^{-10t})(10 \text{ k}) \\&= -0.1 \text{ V}\end{aligned}$$

Thus, the output voltage $v_o(t)$ for $t > 0$ is $\boxed{-0.1 \text{ V}}$.